

LFER Part 2: Substituents with More Punch

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Hammett σ substituent parameters work well in many systems. But in others, there are a set of these parameters that refuse to work well. One thing that these substituents have in common is the ability to accept or donate electrons via resonance. Today we will explore substituent parameters that include these extra resonance effects.

Resonance Parameters

As you read the textbook¹ and try the [suggested problems](#), here are some things that we will be discussing in class.

1. Re-evaluate the Hammett plots for hydrolysis of benzyl chloride, diphenylmethylene chloride and trityl acetates from the previous class meeting² using the Brown-Okamoto substituent parameter, σ^+ . While we are doing this we will consider the following questions. Be prepared to answer them in class.
 - (a) What are the σ^+ substituent parameters? How were they measured? What information do they provide?
 - (b) When would you use σ^+ substituent parameters rather than the Hammett σ substituent parameters?
 - (c) Which plots from the previous meeting were improved when using σ^+ ? Does this fit with your understanding of σ^+ values?
 - (d) Many substituents do not have a unique σ^+ value. What should we use for substituent parameters in these cases?
 - (e) No meta-substituents have unique values for σ^+ . Why?
2. Consider the discussion presented in the following pages³ for the nucleophilic attack of phenoxide ions on epoxides. Be prepared to answer the following questions.
 - (a) The kinetic data seems to correlate better with σ^- values. What are σ^- values? How were they measured? What information do they provide?
 - (b) When would you use σ^- substituent parameters rather than σ or σ^+ parameters?
3. Describe when it is appropriate to use σ , σ^+ or σ^- as the basis for correlating structure with reactivity. Could there be reactions that have only partial sensitivity to resonance effects. Are there parameters that fit in-between σ and σ^+/σ^- ? Consider the discussion at the end of this document about the Yukawa-Tsuno approach⁴ and apply it to the hydrolysis of trityl acetates discussed in the previous meeting.

This document was produced using the $\LaTeX$ typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle* Plots were typeset using *Matplotlib* and *Python* tools. The plot was then further edited in *Affinity Designer*.

¹ Chapter 8.3.5. That's right, this whole class is based on the third decimal place of a textbook chapter. However, you really should read all of ch. 8.2 & 8.3 again.

² See the "Linear Free Energy Relationships, Part 1" handout from the previous meeting for the data tables.

³ See the ["Substituent Constants for Resonance Effects"](#) section.

⁴ See the ["Mix and Match: Partial Resonance Contributions"](#) section

Textbook Questions

As you read about resonance effects and substituent parameters, try these textbook problems: Ch. 8: 24

Learning Goals

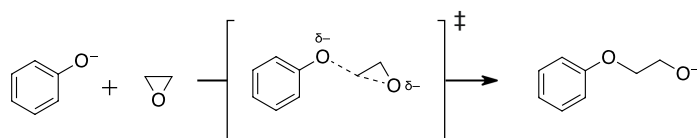
Through the suggested textbook reading, problems and participating in this class discussion, students will...

- have gained further experience in resonance effects and electron delocalization.
- be able to define the parameters in the Brown-Okamoto (σ^+ & σ^-) and the Yukawa-Tsuno LFER systems.
- be able to predict when σ , σ^+ , or σ^- substituent parameters should be used.
- **Computer Skill:** have again used an interactive *Python* notebook to interpret and analyze experimental data from the literature.

Substituent Constants for Resonance Effects

The Hammett equation relates "score" for an electron-withdrawing or electron-donating group to the log of the observed rate or rate constant. The Hammett scores are based on the observed change in the pK_a for substituted benzoic acids. These are designated as σ_m and σ_p for each substituent. Some reactions have electronic effects that are far greater than described in the Hammett σ substituent parameters.

If the reaction centre can connect via resonance with the substituent, there will be a far greater influence by the group. Substituents that are *meta* to the reaction centre cannot connect via resonance, but *para* substituents can. If a group can accept or donate electrons by resonance, we will need to replace Hammett's σ_p values with the σ_p^- and σ_p^+ values of Brown and Okamoto.⁵ I found some data for nucleophilic ring-opening of epoxides by phenoxides in an ancient piece of work⁶ that predates the field of physical organic chemistry⁷ itself.



The oxygen nucleophile of phenoxide would be able to connect via resonance with electron acceptors in the *para* position. Substituents that

⁵ "A Quantitative Treatment for Electrophilic Reactions of Aromatic Derivatives", Y. Okamoto and H.C. Brown, *J. Org. Chem.*, 1957, 22, 485-494. <https://doi.org/10.1021/jo01356a003> and "Substituent Constants for Aromatic Substitution", Herbert C. Brown and Y. Okamoto, *J. Am. Chem. Soc.* 1957, 79, 1913-1917. <https://doi.org/10.1021/ja01565a039>

⁶ "The Velocities of Combination of Sodium Derivatives of Phenols with Olefine Oxides.", David Runciman Boyd and Ernest Robert Marle., *J. Chem. Soc., Trans.*, 1914, 105, 2117-2139. <https://doi.org/10.1039/CT9140502117>

⁷ Louis P. Hammett (1894-1987) did not publish his famous textbook until 1940.

Figure 1: Ring-opening of epoxide by phenoxides

can accept electrons by resonance should have a much greater electron-withdrawing effect compared to the case of benzoic acid. When the electronic structure resembles a phenol, we must use substituent constants designed for the situation.

We will plot the $\log k_{obs}$ vs. σ^- values from data tables. The open circles are points using the Hammett σ_p values and the closed circles are points using σ_m and σ_p^- values. When including Brown-Okamoto parameters in Hammett plots, we generally refer to the axis as σ^+ or σ^- . We use σ_p and σ_m values for substituents where resonance has no significant effect and use the σ_p^+ or σ_p^- values when there is a resonance effect.

Substituent	k_{obs}	σ	σ^+	σ^-
p-NO ₂	0.013	0.78		1.27
p-CN	0.026	0.66		1.00
m-NO ₂	0.032	0.71		
m-CN	0.044	0.56		
p-N ₂ C ₆ H ₅	0.050	0.31	-0.19	0.45
m-Cl	0.101	0.37		
p-Cl	0.108	0.23	0.11	0.19
H	0.205	0		
m-CH ₃	0.256	0.07		
p-CH ₃	0.257	0.17	-0.31	

Table 1: Rate constants for nucleophilic attack of substituted phenols on epoxide.⁶

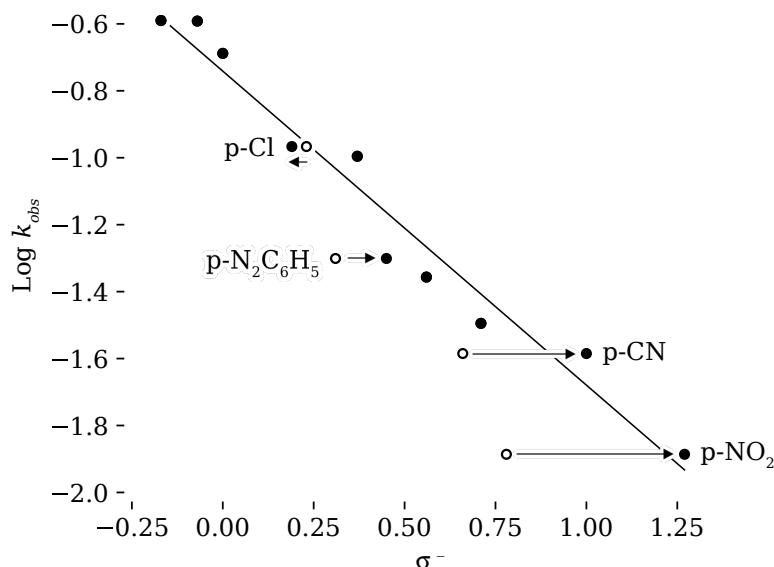


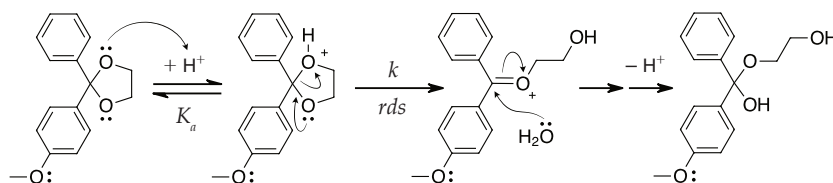
Figure 2: $\log k_{obs}$ vs. σ^- values for ring-opening of epoxide by phenoxides. Open circles are Hammett σ values, closed circles are σ^- values. The slope of the line is -0.90 .

Mix and Match: Partial Resonance Contributions

The Yukawa-Tsuno equation⁸ combines the "scores" for both the mostly inductive substituent constants of Hammett with the resonance substituent constants of Brown & Okamoto. While most Hammett plots fit well when using one set or the other, there are cases where the best fit is when we use a weighted average of the two sets.

Yukawa and Tsuno proposed a plot using a weighting factor, r , that is adjusted to achieve the best line fit. Plots where r is significantly different than 1 or 0 indicate a transition state that has partial resonance participation by the substituent.

The acid hydrolysis of phenylacetals proceeds through a S_N1 mechanism. The protonated leaving group is assisted in departure by the donation of electrons from the adjacent oxygen into the forming p-orbital. Oxygen groups are highly stabilizing for adjacent carbocations and this direct electron donation greatly reduces the importance of the resonance contribution of the Hammett substituents.



We would expect the σ^+ parameter series to fit best when a carbocation is attached to the phenyl ring. However, in this case it seems that neither the σ^+ series or the Hammett σ series fits.⁹ This is because the $p\text{-OCH}_3$ substituent has a large difference between its σ^+ and σ values and neither option lands on the line being telegraphed by the other data points. If only there was a set of σ values for partial resonance donors. That is exactly what Yukawa and Tsuno did. They created a sliding scale between σ and σ^+ using their equation.

The r value represents this scale. When $r = 1$ we have the σ^+ series and when $r = 0$ we have the σ series. In this reaction, the data fits best when we use a series of values in between σ^+ and σ . But how do we find the optimal value for r that gives the best line fit?

Using a literature example,¹⁰ we will explore the Yukawa-Tsuno plot for phenylacetal hydrolysis. The rate data for the hydrolysis reaction is presented in table 2 and substituent constants are presented in table 3.

A Yukawa-Tsuno plot was constructed using the Hammett $\sigma_{m,p}$ constants and the Brown-Okamoto σ^+ constants. See figure 4.

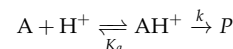
⁸ "Sigma Constants in Electrophilic Reactions and their Interrelation" Y. Yukawa, Y. Tsuno, *Bull. Chem. Soc. Japan*, 1959, 32, 965-971. <https://doi.org/10.1246/bcsj.32.965>

The Yukawa-Tsuno Equation

$$\log \frac{k_X}{k_H} = \rho \cdot [\sigma + r(\sigma^+ - \sigma)]$$

Figure 3: Acid hydrolysis of an acetal

The reaction scheme



The rate law

$$v = k_{obs}[A], k_{obs} = \frac{[H^+]}{K_a + [H^+]}k$$

⁹ The r^2 for the fit with Hammett σ substituent constants is actually good. It is 0.95. Using the σ^+ series of Brown and Okamoto give $r^2 = 0.92$, also an adequate (but not good) fit. using the Yukawa-Tsuno approach we can get $r^2 = 0.97$ (see figure 5.) This is a better fit. All of this is controlled by one single data point, the $p\text{-OCH}_3$ substituent. Does that seem a bit sketchy?

¹⁰ "Substituent Effects in the Acid-Catalyzed Hydrolysis of Cyclic Acetals of Benzophenones", U. Shinya, O. Takumi, N. Toshikazu, *Bull. Chem. Soc. Japan*, 1986, 59, 305-306. <https://www.journal.csj.jp/doi/10.1246/bcsj.59.305>

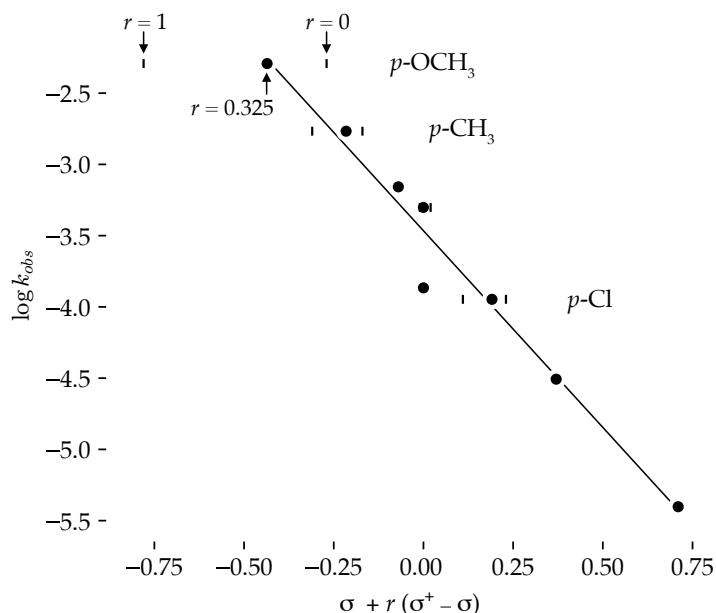


Figure 4: $\log k_{obs}$ vs. σ values for acid hydrolysis of phenylacetals. Vertical slashes represent the edge cases when $r = 0$ (Hammett) and $r = 1$ (Brown-Okamoto). The circular markers represent $r = 0.325$, the slope is $\rho = -2.8$.

Table 2: Rate constants for hydrolysis of substituted diphenylacetals

Substituent	$k_{obs} \times 10^4 \text{s}^{-1}$	$\log k_{obs}$
p-OCH ₃	51.0	-2.29
p-CH ₃	17.1	-2.77
m-CH ₃	6.96	-3.16
p-C ₆ H ₅	4.99	-3.30
H	1.36	-3.87
p-Cl	1.13	-3.95
m-Cl	0.311	-4.51
m-NO ₂	0.0396	-5.40

Table 3: Hammett and Brown-Okamoto substituent constants

Substituent	σ	σ^+	σ^-
p-OCH ₃	-0.27	-0.78	-0.26
p-CH ₃	-0.17	-0.31	-0.17
m-CH ₃	-0.07		
p-C ₆ H ₅	-0.01	0.02	-0.18
H	0.00		
p-Cl	0.23	0.11	0.19
m-Cl	0.37		
m-NO ₂	0.71		

Examine the plot and observe how the two substituents that have the greatest difference between the values of σ and σ^+ fall well to the right or left of the best fit when $r = 0$ and $r = 1$. If we mix the σ and σ^+ values with $r = 0.325$, we observe the best fit as measured by the correlation coefficient r^2 (not to be confused with the Yukawa-Tsuno r factor).¹¹

How do we find the best value for the r factor? One method would be to simply write a program to perform least-squares line fits across a series of values for r . The result for such an approach is shown in figure 5. I then used the `minimize_scalar` function in the `SciPy.stats` library in a *Jupyter* notebook¹² to determine the exact value of r that resulted in the largest value for the correlation coefficient, r^2 . I used $-r^2$ rather than r^2 so that a maximum score would be a minimum in the plot.

Take note that the Yukawa-Tsuno r -value is determined mostly due to the influence of the one or two points on the plot that are most changed as we change r . The r factor is expressed with three significant figures here. Is that justified, given its dependance on a single data point? Also, if your plot does not include substituents with a large variance between the σ and σ^+ constants, then it will not effectively reveal partial resonance effects. Often the Yukawa-Tsuno r -value is set by the p-OCH₃ substituent alone.

¹¹ Throughout this document keep in mind that " r " refers to the Yukawa-Tsuno resonance factor and " r^2 " refers to the Pearson correlation coefficient

¹² See the accompanying *Jupyter* notebook for the documentation of the code and methodology

Figure 5: Optimization of the r -value as scored by the Pearson correlation coefficient, $-r^2$

